NATIONAL UNIVERSITY BANGLADESH



PROVISIONAL

Syllabus
Bachelor of Science (Honours) Second Year
Department of Zoology

Effective from the Academic Year: 2024–2025 and onward

1. **Vision of the University**

We expect that the hundredth anniversary of the National University, our commitment to "Progress for Everyone" will be evident in all areas of our operations and stakeholder engagements.

We leverage our core strength in education, research, and community engagement to address pressing global issues and foster sustainable development.

By 2042, National University aims to strengthen its positions as Bangladesh's top academic institution and rank among the world's leading universities. Our focus will be on expanding expertise building strategic partnerships, and increasing our global influence. Our services will be measured by the achievements of the individuals and communities we serve.

2. **Mission of the University**

To promote "Progress for Everyone" by fostering transformative education, conducting pioneering research, and encouraging meaningful collaboration with individuals, communities, industries, job markets and partners both in Bangladesh and around the world.

3. **Name of the Program**: B.Sc. (Honours) in Zoology

4. **Programme Educational Objectives (PEOs)**

PEO 1: To produce graduates with strong theoretical and practical knowledge of Zoology according to the requirements of contemporary market demand.

PEO 2: To develop graduates capable of performing interdisciplinary and collaborative research and demonstrating technical competence in the field of Zoology.

PEO 3: To develop graduates who can help to deliver adequate, relevant and timely biological knowledge to facilitate research, planning and decision-making program for the government and the community for achieving Sustainable Development Goals (SDGs) of Bangladesh.

PEO 4: To produce graduates with strong leadership, teamwork, communication skills and ethical and moral values that will help them in their professional lives.

5. **Programme Learning Outcomes (PLOs):** At the time of graduation, B.Sc. (Honours) in Zoology students will have achieved the ability to:

PLO 1: Fundamental Knowledge of Zoology: Demonstrate a comprehensive understanding of core zoological concepts, including invertebrate and vertebrate diversity, anatomy, physiology, genetics, evolution, economic importance and ecology.

PLO 2: Taxonomic and Biodiversity Expertise: Identify, classify, and analyze animal species, with emphasis on Bangladesh's native fauna and conservation challenges.

PLO 3: Laboratory and Field Research Skills: Perform advanced laboratory techniques (e.g., microscopy, dissection, staining, basic biochemical tests, DNA analysis) and field studies (e.g., specimen collection, ecological surveys) relevant to zoological research.

PLO 4: Scientific Inquiry and Critical Thinking: Design, conduct, and interpret scientific experiments in Bangladesh's diverse ecosystems using hypothesis-driven approaches and statistical tools.

PLO 5: Conservation and Environmental Awareness: Evaluate climate change and human impacts on wildlife and ecosystems, and propose sustainable conservation strategies for Bangladesh's biodiversity. Assess threats and propose community-based conservation strategies in different situation including vector management and environmental issues.

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PLO 6: Data Analysis and Computational Skills: Utilize bioinformatics, GIS, and statistical software to analyze zoological data and interpret research findings.

PLO 7: Communication and Scientific Reporting: Present research findings effectively through scientific writing, oral presentations, and digital media in both Bengali and English.

PLO 8: Ethical and Professional Responsibility: Follow ethical guidelines in animal handling, research, wildlife conservation, and academic integrity while considering socio-ecological responsibilities.

PLO 9: Interdisciplinary Application: Relate zoology to agriculture, fisheries, public health (e.g., zoonotic diseases), microbiology, and biotechnology for practical applications in Bangladesh.

PLO 10: Career and Leadership Readiness: Develop transferable skills (teamwork, project management, problem-solving) for careers in academia, research, fisheries, wildlife management, public health, and environmental policy.

6. Mapping of PEO with PLO

	PEO-1	PEO-2	PEO-3	PEO-4
PLO-1	√			
PLO-2		√	√	
PLO-3	√	√		
PLO-4		√		√
PLO-5			√	
PLO-6	√	√		√
PLO-7			√	√
PLO-8		√	√	√
PLO-9	√	√	√	
PLO-10	√			√

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7. Courses, Marks and Credit Distribution

SECOND YEAR

Course Code	Course Title	Marks	Credits
8223151	Animal Diversity –II: Lower Chordates and Chordates	100	4
8223153	Comparative Anatomy: Protozoa, Non-chordates and Chordates	100	4
8223155	Environmental Biology	100	4
8223156	Zoology Practical - II	100	4
8222859 8222861	General Chemistry-II Environmental Chemistry	100 50	4 2
8223057 8223058	Botany-II Botany Practical -II	100 50	4 2
8221171	Functional English	100	4
8226601	Applied Information and Communication Technology (Applied ICT)	75	3
8226602	Practical: Applied ICT	25	1
8223160	Viva-Voce (Zoology)	50	2
Total =		**950**	**38**

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Detailed Syllabus

Course Code	8223151	Marks: 100	Credits: 4	Class Hours: 56
Course Title	Animal Diversity –II: Lower Chordates and Chordates			

Course Description

This course provides an in-depth study of the phylum Chordata, tracing evolutionary transitions and structural diversification from protochordates to advanced mammals. Students explore systematic classification up to the order level using diagnostic traits and taxonomic keys. Through detailed comparative anatomy of ten representative type specimens, the course analyzes organ system adaptations across diverse ecological niches. Additionally, students evaluate key evolutionary milestones—including transitional fossils, "living fossils," and primitive taxa—alongside specialized biological phenomena such as avian migration, mammalian radiations, and urgent conservation challenges facing vulnerable species like the Bengal Tiger.

Course Objectives

To enable students to systematically classify, study anatomical, and evaluate the evolutionary and ecological significance of chordate taxa from Urochordata to Mammalia through the study of diagnostic features, type specimens, and specialized groups.

Course Learning Outcomes (CLOs)

After going through the course, students will be able to learn the following outcomes:

CLO1	**Systematic Classification & Diagnosis:** Students will be able to identify and classify chordate taxa (from Protochordata to Mammalia) up to the order level by applying diagnostic morphological characteristics and differentiating between closely related groups using standard taxonomic keys.
CLO2	**Comparative Anatomy & Evolutionary Adaptation:** Students will be able to compare and contrast the skeletal, digestive, circulatory, respiratory, nervous, endocrine and reproductive systems across 10 type study organisms (*Balanoglossus* to* Rabbit*) to explain how anatomical modifications reflect evolutionary adaptations to specific habitats and modes of life.*
CLO3	**Functional Morphology & Life History:** Students will be able to correlate the external morphology, adaptations to habitats, feeding habits, and developmental patterns of type specimens with their ecological niches and behavioral strategies.
CLO4	**Phylogenetic Significance of Transitional Taxa:** Students will be able to evaluate the evolutionary importance of "living fossils" and bridging groups including *Ascidia*, lungfishes, *Latimeria*, *Sphenodon*, and *Archaeopteryx* by analyzing their primitive and derived traits and their position in vertebrate phylogeny.
CLO5	**Zoological Relevance & Conservation Application:** Students will be able to synthesize knowledge of specialized topics (avian migration, mammalian radiations, amphibian parental care) to assess current ecological and conservation challenges facing vulnerable groups such as aquatic mammals, marsupials, flightless birds, and the Bengal Tiger.

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Mapping of CLOs with PLOs

	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10
CLO1	√	√								
CLO2	√									
CLO3	√				√					
CLO4	√									
CLO5					√		√	√		√

Course Contents

Week	Topic	Teaching Learning Strategy	Assessment Strategy	CLOs
1–8 =16 Classes	1. Broad classification of the following up to orders with general and diagnostic characteristics of each taxon with examples and affinities: Hemichordata, Urochordata, Cephalochordata, Myxinae, Cephalaspidomorphi, Chondrichthyes, Osteichthyes, Amphibia, Reptilia, Aves and Mammalia.	Taxonomy Relay → Mystery Specimen Stations→ Character Matrix Construction	• Character Matrix Evaluation • Mystery Specimen Identification Quiz • Formative Class Test	CLO1
9–18 =20 Classes	2. Type study of the following with their systematic position, habitats and external morphology; anatomy including skeletal, digestive, circulatory, respiratory, excretory, nervous, reproductive and endocrine systems; food and feeding habits; mode of life and development: *a.* Hemichordata: *Balanoglossus* *b.* Urochordata: *Herdmania* *c.* Cephalochordata: *Branchiostoma* *d.* Cyclostomata: *Petromyzon* *e.* Chondrichthyes: *Scoliodon* *f.* Osteichthyes: *Labeo rohita/*Tilapia (*Oreochromis*) *g.* Amphibia: *Rana* *h.* Reptilia: *Varanus* *i.* Aves: *Columba livia* j. Mammalia: Rabbit *(Oryctolagus).*	Virtual Dissection Labs (spread across systems) → Expert Jigsaw → Role-Play → Matching Game → Storyboards	• Virtual Dissection Lab Worksheets • Anatomical Storyboard Grading • Mid-Term Examination	CLO1 CLO2 CLO3

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19–28 =20 Classes	**3. Brief notes on the following –** a. Urochordata: *Ascidia* b. Chondrichthyes: Skates and rays. c. Osteichthyes: Lungfishes including *Latimeria*, exotic fishes. d. Amphibia: Bufonids, caecilians, salamanders and newts. e. Reptilia: *Uromastix, Sphenodon*, dinosaurs. f. Aves: *Archaeopteryx*, migration, flightless birds. g. Mammalia: Aquatic mammals, egg-laying mammals, marsupials, apes, Bengal Tiger.	Concept Mapping → Debate Panel → Migration Simulation → Case Study Analysis → Conservation Pitch (final project)	• Concept Mapping Evaluation • Debate Participation & Case Study Rubric • Conservation Pitch Final Project • Semester Final Examination	CLO4 CLO5

Course Assessment Framework:

Total Course Marks: 100

Assessment Category	Assessment Component	Weightage
Formative Assessment (20%)	Attendance & Class Participation	5%
	In-course Examination	10%
	Quiz / Assignment Presentation	5%
Summative Assessment (80%)	Final Examination • Students will have to answer 8 questions out of 12 • 10 marks per question ($8 \times 10 = 80$ marks) • Each question may contain maximum 3 sub-sections (e.g., a, b, c)	80%

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List of Books

Benton, M. J. (2014). *Vertebrate Palaeontology* (4th ed.). Wiley-Blackwell.

Hickman, C. P., Jr., Keen, S. L., Eisenhour, D. J., Larson, A., & I'Anson, H. (2024). *Integrated Principles of Zoology* (19th ed., International student ed.). McGraw Hill.

Hilderbrand, M., & Goslow, G. E. (2001). *Analysis of Vertebrate Structure* (5th ed.). John Wiley & Sons.

Jordan, E. L., & Verma, P. S. (2022). *Chordate Zoology* (14th ed.). S. Chand and Company.

Kardong, K. V. (2019). *Vertebrates: Comparative Anatomy, Function, Evolution* (8th ed.). McGraw-Hill Education.

Kent, G. C., & Carr, R. K. (2000). *Comparative Anatomy of the Vertebrates* (9th ed.). The McGraw-Hill Companies.

Liem, K. F., Bemis, W. E., Walker, W. F., & Grande, L. (2001). *Functional Anatomy of the Vertebrates: An Evolutionary Perspective* (3rd ed.). Harcourt College Publishers.

Moorthi, M., Senthil Kumar, A., & Govindarajan, M. (2024). *Practical Zoology (Invertebrates & Chordates)*. LAP Lambert Academic Publishing.

Pough, F. H., Janis, C. M., & Heiser, J. B. (2013). *Vertebrate Life* (9th ed.). Pearson.

Weichert, C. K., & Presch, W. (1970). *Elements of Chordate Anatomy*. Tata McGraw-Hill.

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Course Code	8223153	Marks: 100	Credits: 4	Class Hours: 56
Course Title	Comparative Anatomy: Protozoa, Non-chordates and Chordates			

Course Description

This course provides a comprehensive study of the structural and functional diversity of animals across the evolutionary continuum from Protozoa to Mammalia. Students will explore the systematic classification, morphological complexity, and specialized life strategies of major non-chordate phyla alongside the comparative anatomy of chordate lineages. Emphasis is placed on the evolutionary progression of key organ systems—including integumentary, skeletal, digestive, cardiorespiratory, nervous, and urinogenital systems—and major anatomical milestones such as coelom development, cephalization, and terrestrial adaptations. Through comparative analysis, students will connect structural adaptations to ecological roles and contemporary biodiversity conservation challenges.

Course Objectives

To enable students to comprehensively understand the structural and functional diversity of animals from Protozoa to Mammalia, emphasizing comparative anatomy, evolutionary adaptations, and systematic classification, while developing skills to analyze organ system evolution, ecological significance, and conservation relevance of major taxa.

Course Learning Outcome (CLO)

After going through the course, students will be able to learn the following outcomes:

CLO1	Classify and differentiate major non-chordate phyla (Protozoa to Echinodermata) based on their structural complexity, locomotory mechanisms, reproductive strategies, and specialized adaptations (e.g., parasitism, polymorphism, metamerism).
CLO2	Analyze the evolutionary progression of key organ systems including, integument, skeletal, digestive, respiratory, circulatory, nervous, and urinogenital systems from non-chordates to mammals, explaining structural modifications in response to adaptive pressures.
CLO3	Compare and contrast the structural organization and functional anatomy of chordate taxa (Integument to Urinogenital) to evaluate evolutionary innovations such as jaw suspension, amniotic adaptations, flight adaptations, and mammalian radiations.
CLO4	Evaluate the evolutionary significance of major morphological innovations including, coelom, metamerism, cephalization, jaw suspension, kidney evolution, and neural complexity using comparative anatomical evidence from both non-chordate and chordate lineages.
CLO5	Synthesize knowledge of vertebrate diversity and adaptations (e.g., flight, aquatic life, terrestrial colonization) to assess ecological and conservation challenges, applying comparative zoological principles to real-world biodiversity issues.

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Mapping of CLOs with PLOs

	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10
CLO1	√	√								
CLO2	√									
CLO3	√	√								
CLO4	√			√						
CLO5		√			√		√	√		√

Course Contents

Weeks	Topic	Teaching Learning Strategy	Assessment Strategy	CLOs
1-14 = 28 Classes	**Protozoa & Non-Chordates (50 marks)** 1. **Protozoa:** Structural complexity of free-living vs. parasitic forms; Locomotory organelles and locomotion, reproduction, parasitism; Nuclear apparatus and nutrition. 2. **Porifera: S**keletal structures and canal systems. 3. **Coelenterata:** Polymorphism and metagenesis; coral reef, reef formation and theories. 4. **Platyhelminthes:** Parasitic adaptations, holdfast organs. 5. **Annelida:** Muscular contraction; Metamerism, modifications in nephridia and chaetae; Asexual reproduction of Polychaeta, segmental organs. 6. **Arthropoda:** Crustacean larvae, respiratory organs (gills, trachea, book lungs, etc.) 7. **Mollusca:** Structure, formation and modification of Shell, mantle and foot. 8. **Echinodermata:** Symmetry of Echinoderms; Larval forms. 9. Evolution of the digestive tract, coelom, metamerism and cephalization; Evolutionary	Lecture, Assignment on Parasitic Adaptation, Comparison Chart Dissection, Photograph/ Diagrams Analysis among "Breathing in Arthropods" Jigsaw Shell Morphology Gallery "Coral Reef Ecosystem" Concept Mapping	• Formative Quizzes • Assignment Evaluation (Comparison Charts & Concept Maps) • Active Participation in Jigsaw & Gallery Tasks • Mid-Term Examination (Non-Chordates)	CLO1 CLO2 CLO4

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	significance of non-chordate larval forms;			
15-28 = 28 Classes	**Chordates (50 Marks)** 1. **Integument and its derivatives:** Structure of skin**,** Integumentary glands and outgrowths, including fins, scales, feathers, hairs, horns, hooves, claws, beaks and nails. 2. **Skeletal system:** Structure and function of the vertebral column, jaw suspension, appendicular skeleton (girdles and limbs) and cranial anatomy. 3. **Digestive system:** General organization and anatomy of gut; tongue, mouth, and teeth structure; dentition, dental formula; evolution of the liver and pancreatic glands. 4. **Respiratory and Circulatory system:** Evolution from gills to lung; accessory respiratory organs; modification of aortic arches and heart in reptiles, birds and mammals. 5. **Nervous system:** Evolution of the brain, cranial nerves and special sensory structures. **6. Urinogenital system:** Evolution of kidney types and reproductive ducts. **7. Vertebrate diversity and adaptations:** Bird's flight adaptations and feather structure; Early tetrapod and structural adaptations for dual life of Amphibians; Amniote adaptations and skull types of Reptiles; Distinctive mammalian features (hair, mammary glands) and adaptive radiations.	Lecture, Group Discussion, Assignment on comparative skin, Histology station, Kidney, Evolution flowcharts, Mammalian adaptations" Gallery Walk, Feather morphology lab, Skull Typology Workshop	• Station Activity Reports / Lab Worksheets •Group Discussion & Presentation Assessment •Flowchart & Portfolio Review • Semester Final Examination (Comprehensive & Chordates)	CLO2 CLO3 CLO4 CLO5

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Course Assessment Framework:

Total Course Marks: 100

Assessment Category	Assessment Component	Weightage
Formative Assessment (20%)	Attendance & Class Participation	5%
	In-course Examination	10%
	Quiz / Assignment Presentation	5%
Summative Assessment (80%)	Final Examination • Students will have to answer 8 questions out of 12 • 10 marks per question ($8 \times 10 = 80$ marks) • Each question may contain maximum 3 sub-sections (e.g., a, b, c)	80%

List of Books

Benton, M. J. (2014). *Vertebrate Palaeontology* (4th ed.). Wiley-Blackwell.

Hickman, C. P., Jr., Keen, S. L., Eisenhour, D. J., Larson, A., & I'Anson, H. (2024). *Integrated Principles of Zoology* (19th ed.). McGraw Hill.

Hilderbrand, M., & Goslow, G. E. (2001). *Analysis of Vertebrate Structure* (5th ed.). John Wiley & Sons.

Jordan, E. L., & Verma, P. S. (2022). *Chordate Zoology* (14th ed.). S. Chand and Company.

Kardong, K. V. (2019). *Vertebrates: Comparative Anatomy, Function, Evolution* (8th ed.). McGraw-Hill Education.

Kotpal, R. L. (2021). *Modern Textbook of Zoology: Invertebrates* (12th ed.). Rastogi Publications.

Liem, K. F., Bemis, W. E., Walker, W. F., & Grande, L. (2001). *Functional Anatomy of the Vertebrates: An Evolutionary Perspective* (3rd ed.). Harcourt College Publishers.

Pough, F. H., Janis, C. M., & Heiser, J. B. (2013). *Vertebrate Life* (9th ed.). Pearson.

Ruppert, E. E., Fox, R. S., & Barnes, R. D. (2004). *Invertebrate Zoology: A Functional Evolutionary Approach* (7th ed.). Brooks/Cole.

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Course Code	8223155	Marks: 100	Credits: 4	Class Hours: 56
Course Title	Environmental Biology			

Course Description

This course explores the fundamental principles of environmental science, covering ecosystem dynamics, biodiversity, and pollution mechanisms across air, water, and soil. Students examine ecotoxicology, climate change impacts, and analytical assessment tools like EIA and GIS alongside waste management and microbial remediation. The course emphasizes environmental laws, disaster risk mitigation, and sustainable conservation strategies, with a special focus on Bangladesh.

Course Objectives

To provide a comprehensive understanding of environmental science, encompassing ecosystem components, biodiversity, pollution dynamics, climate change, and environmental management. This course aims to equip students to assess environmental impacts, evaluate ecological threats, and propose sustainable solutions to environmental challenges, with particular emphasis on Bangladesh's context.

Course Learning Outcomes (CLOs)

After going through the course, students will be able to learn the following outcomes:

CLO1	Identify and classify the components of the environment (atmosphere, hydrosphere, lithosphere, biosphere), biodiversity, and ecological processes, and explain their interrelationships in maintaining ecosystem balance.
CLO2	Analyze the sources, mechanisms, and effects of various environmental pollutants (air, water, soil, and sound) and evaluate their impact on ecosystems and human health using physicochemical and ecotoxicological parameters (DO, BOD, COD, bioaccumulation, biomagnification).
CLO3	Evaluate the causes and consequences of climate change phenomena including, greenhouse effect, global warming, ozone depletion, ocean acidification, and acid rain and assess their specific impacts on Bangladesh's ecosystems and communities.
CLO4	Apply environmental assessment tools (EIA, GIS, remote sensing), microbial degradation principles, and waste management strategies to propose sustainable solutions for pollution control, disaster management, and conservation in Bangladesh.
CLO5	Synthesize knowledge of environmental policies, laws, and sustainable development frameworks to formulate evidence-based management plans addressing disaster vulnerability, energy resource management, and conservation challenges specific to Bangladesh.

Mapping of CLOs with PLOs

	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10
CLO1	√									
CLO2	√	√			√				√	
CLO3	√				√					
CLO4			√		√	√			√	
CLO5					√		√	√		√

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Course Contents

Week	Topic	Teaching Learning Strategy	Assessment Strategy	CLOs
1–3=6 Classes	**1. Environment:** Structure, composition and thermal stratification of atmosphere, hydrosphere, lithosphere, and biosphere; structural components of ecosystems, ecological efficiencies and system productivity.	Lectures and Group Discussions Concept Mapping	• Class Participation • Concept Map Evaluation	CLO1
4–6=6 Classes	**2. Biodiversity:** Biodiversity frameworks (α, β & γ diversity); global biodiversity hotspots; major drivers of faunal loss; biodiversity assessment and sampling techniques.	Biodiversity Sampling Simulation In-situ vs. Ex-situ Debate	• Simulation Worksheet • Debate Rubric Assessment	
7–10=8 Classes	**3. Ecotoxicology:** Absorption, distribution, metabolism, and excretion (ADME) pathways of environmental toxins in living organisms; genotoxicity, mutagenicity, endocrine disruption, and biomarkers of oxidative stress; hazard identification, dose-response assessment, exposure assessment, and risk characterization; bioaccumulation, biomagnification, toxicological testing.		• Practical and Lab Report • Data Calculation (BOD/COD)	
11–14=8 Classes	**4. Environmental Pollution:** Sources, effects and management of air (PM-10, PM-2.5, VOCs), photochemical smog mechanisms; effect of climate change on ecosystems, range shifts of species, and carbon management; impacts of xenobiotics on fauna and ecosystem; solid waste management.		• Mid-Term Examination • Wastewater Treatment Case Review	
15–18 =8 Classes	**5. Soil Biology and Management:** Physical and chemical properties of soil and their control over microbial distribution; functional soil spheres; soil organic matter dynamics; soil microflora and fauna; soil food web interactions; drivers of soil degradation; sustainable soil management practices.	Microbial Degradation Lab GIS Mapping Workshop	• Lab Notebook Evaluation • Waste Management Flowchart Rubric	CLO4
19–21 =6 Classes	**6. Tools for Environmental Analysis:** Remote sensing and Geographical Information System (GIS) for habitat mapping; Chromatography Systems; Spectroscopy.	Microbial Degradation Lab GIS Mapping Workshop	• GIS Map Output Assessment • Practical Task Review	
22–24 =6 Classes	**7. Disaster Management:** Earthquake, flood, cyclone, tidal surges, tsunami, saline water		• Presentation Evaluation • Bangladesh	

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	intrusion, drought, river erosion and siltation with particular reference to Bangladesh.		Vulnerability Report	
25–28 =8 Classes	**8. Environmental Policies, Laws, and Ethics:** National and international acts on water, air, wildlife and environment protection; steps, screening, scoping, and public hearing on environmental impact assessment; the ethical implications of developed nations exporting electronic waste, plastics, and toxic materials to developing nations; the moral obligations of the present generation to safeguard climate stability and natural resources for future generations.	Lecture, Group Discussion, and Assignment	• EIA Term Paper / Assignment Evaluation • Semester Final Examination	CLO5

Course Assessment Framework:

Total Course Marks: 100

Assessment Category	Assessment Component	Weightage
Formative Assessment (20%)	Attendance & Class Participation	5%
	In-course Examination	10%
	Quiz / Assignment Presentation	5%
Summative Assessment (80%)	Final Examination • Students will have to answer 8 questions out of 12 • 10 marks per question ($8 \times 10 = 80$ marks) • Each question may contain maximum 3 sub-sections (e.g., a, b, c)	80%

List of Books

Ahmed, A. U., & Haque, N. (2023). *Climate Change and Disaster Management in Bangladesh: A Comprehensive Review*. Bangladesh Centre for Advanced Studies (BCAS).

Alam, M., & Hossain, M. S. (2020). *Environmental Challenges and Sustainable Development in Bangladesh: Science, Policy and Practice*. Springer.

Campbell, J. B., & Wynne, R. H. (2011). *Introduction to Remote Sensing* (5th ed.). Guilford Press.

Cunningham, W. P., & Cunningham, M. A. (2021). *Principles of Environmental Science: Inquiry and Applications* (10th ed.). McGraw-Hill Education.

Manahan, S. E. (2017). *Environmental Chemistry* (11th ed.). CRC Press.

Miller, G. T., & Spoolman, S. E. (2023). *Living in the Environment* (20th ed.). Cengage Learning.

Pepper, I. L., Gerba, C. P., & Gentry, T. J. (2015). *Environmental Microbiology* (3rd ed.). Academic Press.

Wright, R. T., & Boorse, D. F. (2017). *Environmental Science: Toward a Sustainable Future* (13th ed.). Pearson.

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Course Code	8223156	Marks: 100	Credits: 4	Class Hours: 56
Course Title	Zoology Practical-II			

1. Study of museum specimens: Representatives of all types of chordates with reference to Bangladesh (at least 50 specimens to be studied).

Herdmania pallida, Ascidia mentula, Salpa maxima, Doliolum denticulatum, Branchiostoma lanceolatum, Scoliodon laticaudus, Labeo rohita, Labeo calbasu, Catla catla, Cirrhinus mrigala, Channa striata, Channa punctata, Channa marulius, Clarias batrachus, Heteropneustes fossilis, Anabas testudineus, Wallago attu, Tenualosa ilisha, Puntius sophore, Notopterus notopterus, Mystus vittatus, Setipinna phasa, Chitala chitala, Pangasius pangasius, Eutropiichthys vacha, Ompok pabda, Mystus aor, Mastacembelus armatus, Chanda nama, Monopterus cuchia, Labeo bata, Euphlyctis cyanophlyctis, Bufo melanostictus, Microhyla ornata, Polypedates leucomystax, Rana tigrina, Naja kaouthia, Bungarus caeruleus, Daboia russelii, Xenochrophis piscator, Naja naja, Ophiophagus hannah, Python molurus, Calotes versicolor, Varanus bengalensis, Bungarus fasciatus, Crocodylus palustris, Copsychus saularis, Corvus splendens, Acridotheres tristis, Passer domesticus, Halcyon smyrnensis, Ardeola grayii, Egretta garzetta, Psittacula krameria, Spilornis cheela, Pycnonotus cafer, Columba livia, Dinopium benghalense, Macaca mulatta, Axis axis, Felis chaus, Platanista gangetica.

2. Study of bones: Comparative study of the skeleton of an amphibian, a reptile, a bird, and a mammal.

3. Dissection: Dissection of the following specimens:

Major dissection:

a) Any fish – Brain and cranial nerves

b) Pigeon/*Bufo* – Circulatory system

c) Pigeon – Air sacs

Minor dissection:

1. Lata fish- Afferent/efferent blood vessels
2. Eye muscle of any fish.

4. Temporary mounting: Scales and Weberian ossicles of fishes.

5. Freshwater studies: Sampling and identification of plankton and benthic microfauna in freshwater samples.

6. Estimation of soil parameters, water quality testing, and identification of local flora/fauna.

7. Study of permanent slides: Whole mount, embryological stages, and histological slides (at least 20 slides to be studied).

8. Generate geographical map by using GPS reading

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Distribution of Marks for Second Year Final Practical Examination

No.	Particulars			Marks
1.	Major dissection (dissection 7 + display 2 + drawing and labeling 3)			12
2.	Minor dissection (dissection 4 + display 2 + drawing and labeling 2)			8
3.	Temporary mounting (staining 1+ mounting 2+ drawing 2 + labeling 1)			6
4.	Spotting of museum specimens			
	a.	Vertebrate specimens from different classes	8 items × 2 = 16 marks.	
	b.	Permanent slides	3 items × 2 = 6 marks.	
	c.	Bones	3 items × 2 = 6 marks.	
	14 items × 2			28
5.	Freshwater studies (5 species of microfauna to be shown – identification and classification 2, characteristics 1) – 5 items × 2 =			10
6.	Conduct field study at terrestrial/aquatic habitats, identify the flora/fauna by quadrat surveying and prepare a scientific report (Experiment-4, Title-1, Abstract-2, Introduction-2, Objectives-1, Materials and Methods-1, Results-2, Discussions-1, Acknowledgement-1, and References with APA style-1)			16
7.	Prepare a geographical map based on GIS reading of different objects to your campus or a study area by showing mosquito breeding sites/drains/water bodies/trees/animal living habitats/etc.)			10
8.	Practical Note Book =			10
	Total			100

List of Books

Ekaratne, S. U. K., & Dissanayake, D. M. S. B. (2019). *A Manual of Practical Zoology: Chordates*. CBS Publishers & Distributors.

Ghosh, A. K., & Chakraborty, S. (2021). *A Laboratory Manual of Vertebrate Zoology* (3rd ed.). New Central Book Agency.

Kohli, M. P. S., & Khanna, D. R. (2018). *Practical Zoology: Vertebrates*. Discovery Publishing House.

Lal, S. S. (2022). *Practical Zoology: Vertebrates* (4th ed.). Rastogi Publications. (Note: Covers dissection, bones, mounts, and museum specimen studies).

Verma, P. S. (2020). *A Manual of Practical Zoology: Chordata*. S. Chand and Company.

PROVISIONAL

Course Code	8222859	Marks: 100	Credits: 4	Class Durations: 28 Weeks
Course Title:	General Chemistry-II			

Course Description

This course provides students with a comprehensive understanding of fundamental concepts and principles of general chemistry, including the chemistry of nonmetals and metals, thermodynamics, chemical kinetics, electrochemistry, catalysis, solid-state chemistry, coordination chemistry, aromatic and organic reactions, carbohydrates, amino acids, and polymers. The course emphasizes the understanding of chemical principles, structures, reactions, mechanisms, properties, and applications across inorganic, physical, organic, and coordination chemistry.

Course Objectives

Upon completion of this course, students will be able to:

1. Explain the fundamental properties and chemical behaviour of nonmetals, metals, transition metals, and inner transition metals.
2. Explain the principles of thermodynamics, energy changes, chemical kinetics, electrochemistry, and catalysis.
3. Describe the structures, properties, and characteristics of crystalline and amorphous solids and coordination compounds.
4. Explain aromaticity, heterocyclic compounds, major organic reactions, and their mechanisms and applications.
5. Understand the structures, classification, properties, and reactions of carbohydrates, amino acids, and polymers.
6. Apply fundamental chemical concepts to explain chemical properties, reactions, mechanisms, and related phenomena.

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Course Learning Outcomes (CLOs)

CLO	Course Learning Outcome	Bloom's Level
CLO1	Explain the general properties, structures, bonding, electronic characteristics, and chemical behaviour of nonmetals, metals, transition metals, and inner transition metals.	Understand
CLO2	Explain and apply fundamental principles of thermodynamics, energy changes in chemical reactions, chemical kinetics, electrochemistry, and catalysis.	Apply
CLO3	Describe the properties and structures of solids and explain fundamental concepts of coordination chemistry, including coordination compounds, ligands, nomenclature, bonding, and stability.	Understand
CLO4	Explain aromaticity, aromatic electrophilic substitution reactions, heterocyclic compounds, and major organic reactions with their mechanisms and applications.	Analyze
CLO5	Describe the definition, classification, structures, properties, and reactions of carbohydrates and amino acids and explain the basic concepts and classifications of polymers.	Understand
CLO6	Analyze chemical concepts, reactions, structures, mechanisms, and properties across inorganic, physical, organic, and coordination chemistry.	Analyze

Mapping CLOs with PLOs

CLO / PLO	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	✓	✓	✓	–	–	✓	–	✓	✓	✓	–	–
CLO2	✓	✓	✓	–	✓	✓	–	✓	✓	✓	–	✓
CLO3	✓	✓	✓	–	–	✓	–	✓	✓	✓	–	–
CLO4	✓	✓	✓	✓	–	✓	–	✓	✓	✓	–	✓
CLO5	✓	✓	✓	–	–	✓	–	✓	✓	✓	–	–
CLO6	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓

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Course Contents

Module No.	Contents	Teaching–Learning Strategies	Assessment Methods	CLOs
1	**Nonmetals:** General properties of nonmetals, ortho and para hydrogen molecules, allotropy of carbon, catenation, halogens and their basic properties, chemistry of noble gases.	Lecture, interactive discussion, molecular representation	Written examination, class test, short questions	CLO1
2	**Metals:** Metallic bond, electron sea theory of metallic bond, characteristics of metals, band theory of conductivity, conductors, semiconductors and insulators, transition metals and inner transition metals colour and magnetism in transition metal chemistry.	Lecture, discussion, diagrams, conceptual problem-solving	Written examination, class test, short questions	CLO1, CLO6
3	**Energy changes in chemical Reactions:** System and surroundings, open system and closed system, thermodynamics, state functions, the first law of thermodynamics, the concept of internal energy and enthalpy, measurement of enthalpy changes, enthalpy of formation, Hess's law, lattice enthalpy, Born-Haber cycle, second law of thermodynamics, entropy and free energy.	Lecture, numerical problem-solving, board work, discussion	Numerical problems, written examination, class test	CLO2, CLO6
4	**Rates of chemical Reactions:** Reaction rate, rate constant, rate law, order of reactions, first order reaction, half-life, order and molecularity, effect of temperature on the rate of reaction, collision theory and reaction rates, activation energy, Arrhenius equation.	Lecture, numerical exercises, graphical explanation, problem-solving	Numerical problems, written examination, class test	CLO2, CLO6
5	**Electrochemistry:** Redox reactions, electrolytic and galvanic cells, cell notation, standard reduction potentials, emf of cells, the effect of concentration	Lecture, cell-diagram demonstration, problem-solving,	Written examination, numerical problems, class	CLO2, CLO6

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	of cell emf, batteries, corrosion.	discussion	test	
6	**Catalysis:** Catalyst, homogeneous catalysis, enzyme catalysis, auto catalysis.	Lecture, discussion, examples, conceptual explanation	Written examination, short questions, class test	CLO2
7	**Solids:** Properties of solids, crystalline and amorphous solids, distinction between crystalline and amorphous solids, isomorphism, polymorphism and allotropy, crystal lattice unit cell crystal systems Bragg's law.	Lecture, diagrams, model/visual demonstration, discussion	Written examination, short questions, problem-solving	CLO3, CLO6
8	**Coordination Chemistry:** Coordination compounds, ligands, coordination number, nomenclature, structures of complex compounds, Werner's primary and secondary valency concept, sidwick's electronic concept, valence bond theory, stability of coordination compounds.	Lecture, structural representation, nomenclature exercises, discussion	Written examination, nomenclature exercises, class test	CLO3, CLO6
9	**Aromatic Compounds:** Aromaticity aromaticity of benzene, Electrophillic aromatic substitution reactions with reference to nitration halogention, sulphonation and alkylation. Heterocyclic compounds: Pyrrole, furan, thiophene, pyridine.	Lecture, structural representation, reaction mechanism discussion, problem-solving	Written examination, reaction-based questions, class test	CLO4
10	**Organic reactions:** Brief study on Electrophilic addition, Nucleophilic addition, Elimination reaction, condensation reaction, oxidation, and reduction reactions and organic compounds. Mechanism and application of the following reactions, Friedel Craft reaction, Clemmenson reduction, Wolf Krishner reduction, Perkin reaction, Claisen reaction, Cannizzaro reaction and Aldol	Lecture, reaction mechanism demonstration, structural representation, problem-solving	Written examination, reaction mechanisms, class test	CLO4, CLO6

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	condensation.			
11	**Carbohydrates:** Definition, classification, structure and reactions of monosaccharides. Polysaccharide-cellulose and starch.	Lecture, structural representation, discussion, reaction-based learning	Written examination, short questions, class test	CLO5
12	**Amino Acids:** Structures classification, synthesis physical and chemical properties of amino acids.	Lecture, structural representation, discussion, problem-solving	Written examination, short questions, class test	CLO5
13	**Polymer Chemistry:** Polymers homopolymer, heteropolymer, low density and high-density polymer, copolymers, studies of some polymers-polyvinylchloride, nylon 66, silk and wool.	Lecture, classification charts, discussion, examples	Written examination, short questions, class test	CLO5

Course Assessment Framework

Total Course Marks: 100

Assessment Category	Assessment Component	Weightage
Formative Assessment (20%)	Attendance & Class Participation	5%
	Incourse Examination	10%
	Quiz / Assignment Presentation	5%
Summative Assessment (80%)	Final Examination • Students will have to answer 8 questions out of 12 • 10 marks per question ($8 \times 10 = 80$ marks) • Each question may contain maximum 3 sub-sections (e.g., a, b, c)	80%

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List of Books

Cotton, F. A., Wilkinson, G., & Gaus, P. L. (1987). *Basic inorganic chemistry*. John Wiley & Sons.

Ebbing, D. D., & Wrighton, M. S. (1996). *General chemistry*. Houghton Mifflin.

Finar, I. L. (1973). *Organic chemistry: Volume I: The fundamental principles*. Longmans.

Haider, S. Z. (n.d.-a). *Introduction to modern inorganic chemistry*. Friends International.

Haider, S. Z. (n.d.-b). *Selected topics on advanced inorganic chemistry*. Students' Publication.

Huque, M. M., & Nawab, M. A. (n.d.). *Principles of physical chemistry*. Students' Publications.

Madan, R. D. (n.d.). *Modern inorganic chemistry*. S. Chand & Company Ltd.

Malik, W. U., Tuli, G. D., & Madan, R. D. (n.d.). *Selected topics in inorganic chemistry*. S. Chand & Company Ltd.

Morrison, R. T., & Boyd, R. N. (n.d.). *Organic chemistry*.

Silberberg, M. (2009). *Chemistry: The molecular nature of matter and change*. McGraw-Hill Higher Education.

PROVISIONAL

Course Code	8222861	Marks: 50	Credits: 2	Class Hours: 28
Course Title	Environmental Chemistry			

Course Description

This course provides a comprehensive introduction to the chemical principles governing the natural environment and the impact of anthropogenic activities on earth systems. It encompasses a detailed study of the structural segments of the environment atmosphere, hydrosphere, lithosphere, and biosphere and focuses on the chemical pathways, transport dynamics, and transformations of pollutants. Students will explore the mechanisms behind critical global phenomena, including acid rain, the greenhouse effect, ozone depletion, and bio-magnification. The curriculum emphasizes quantitative water quality metrics (pH, DO, BOD, COD), modern wastewater treatment processes, soil degradation vectors, and the biochemical toxicology of toxic persistent compounds like PCBs and heavy metals. Through theoretical frameworks, case studies, and numerical assessments, this course equips students with the essential analytical tools needed to evaluate environmental degradation and formulate sustainable mitigation strategies.

Course Objectives

Upon successful completion of this course, students will be able to:

1. Develop a fundamental understanding of the chemical composition, interactions, and structural layers defining environmental segments.
2. Analyze the chemical reactions, source identification methods, and transport pathways of hazardous air, water, and soil pollutants.
3. Evaluate industrial and municipal water contamination by measuring key water quality parameters and understanding standard purification technologies.
4. Explain the biochemical mechanisms of toxicity, bioaccumulation, and environmental hazards associated with heavy metals and persistent organic compounds.
5. Apply physical and chemical principles to design sustainable control mechanisms for industrial effluents, urban waste management, and soil remediation.
6. Cultivate scientific reasoning, problem-solving abilities, and an ethical foundation required to address domestic and global environmental challenges.

Course Learning Outcomes (CLOs)

CLO	Course Learning Outcome	Bloom's Level
CLO 1	Define and explain the fundamental components of the environment and their systemic interactions.	Understand
CLO 2	Analyze the sources, chemical behavior, transport, and environmental consequences of air, water, and soil pollutants.	Analyze
CLO 3	Evaluate key water and waste water quality parameter sand interpret their environmental significance.	Evaluate

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CLO 4	Assess the biochemical toxicity, environmental impacts, and control methods of heavy metals and hazardous compounds such as PCBs.	Evaluate

Mapping CLOs to PLO Matrix

CLO	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	✓						✓			✓		
CLO2		✓					✓					
CLO3			✓				✓					
CLO4							✓					

Course Contents

The original topics have been chronologically balanced across the 28 class hours within 28 weeks using the layout patterns validated in the curriculum text:

Week	Topics	Teaching-Learning Strategy	Assessment Strategy	CLO
Week 1-3	**1. Environment:** Introduction components of environment, factors affecting environment, environmental management, environment and health, environmental chemistry, segments of environment – atmosphere, hydrosphere, lithosphere and biosphere, structure of atmosphere.	Interactive Lecture, Group Discussion, Concept Mapping	Quiz, Class Test, Short Assignment	**CLO1**
Week 4-7	**2. Pollution and Pollutants:** Pollution, environmental pollution, pollutant, classification of pollutants, types of pollution. PCBs and their sources and hazards, Detection & estimation of PCBs. Bio-magnification.	Lecture, Case Study Analysis, Group Discussion	Class Test, Assignment, Viva-voce	**CLO2 CLO4**
Week 8-14	**3. Air Pollution:** Introduction, air quality, major sources of air pollution, gaseous pollutants. Acid rain-how acid rain is formed, adverse effects of acid rain. Greenhouse effect-how the greenhouse effect is produced, consequences of greenhouse effect and global warming. El Niño phenomenon and its effect. Ozone depletion, mechanism of ozone depletion, effects of ozone depletion.	Lecture, Multimedia Presentation, Problem-Based Learning	Quiz, Mid Term Exam, Assignment	**CLO2**
Week 15-19	**4. Water Pollution:** Introduction, classification of water pollutants. Physical,	Interactive Lecture,	Class Test, BOD/COD	**CLO3**

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	chemical and biological characteristics of wastewater. Industrial wastewater treatment, municipal water treatment. Water quality parameters and standards, measurements of important parameters such as pH, DO, BOD, COD and temperature for water quality assessments.	Numerical Exercises, Laboratory Demonstration	Problems, Report Writing	
Week 20-24	**5. Soil Pollution:** Composition of soil, importance of soil to the biosphere, sources of soil pollution. Effects of soil pollution-synthetic fertilizer and pesticides, effects of industrial effluents, effects of urban wastes, control of soil pollution.	Lecture, Case Study Analysis, Group Discussion	Quiz, Assignment	**CLO2**
Week 25-28	**6. Heavy metals in the Environment:** Trace metals, light metals and heavy metals, deadly heavy metals, sources of heavy metals, biochemical effects, toxicity, toxicology, control and treatment of mercury, chromium, arsenic and lead.	Lecture, Problem-Based Learning, Course Revision	Final Assessment, Viva-voce	**CLO4**

Course Assessment Framework

Total Course Marks: 50

Assessment Category	Assessment Component	Weightage
Formative Assessment (20%)	Attendance & Class Participation	4%
	Incourse Examination	10%
	Quiz / Assignment Presentation	6%
Summative Assessment (80%)	Final Examination • Students will have to answer 4 questions out of 6 • 10 marks per question ($4 \times 10 = 40$ marks) • Each question may contain maximum 3 sub-sections (e.g., a, b, c)	80%

List of Books

Dara, S. S., & Mishra, D. D. (2006). *A textbook of environmental chemistry and pollution control*. S. Chand Publishing.

De, A. K. (2017). *Environmental chemistry* (8th ed.). New Age International Publishers.

Manahan, S. E. (2010). *Environmental chemistry* (9th ed.). CRC Press.

Sharma, B. K., & Kaur, H. (2018). *Environmental chemistry* (16th ed.). Goel Publishing House.

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Course Code	8223057	Marks:100	Credits: 4	Class Hours: 56
Course Title	Botany - II			

Course Description

This course provides an integrated understanding of fundamental concepts in cell biology, genetics, plant breeding, biotechnology, ecology, conservation biology, plant physiology, and phytochemistry. It examines cellular organization and division, inheritance and genetic variation, plant improvement and evolution, biotechnological applications, ecosystem structure and function, biodiversity conservation, plant physiological processes, and major plant biomolecules. The course emphasizes the application of botanical knowledge to plant growth, crop improvement, environmental management, biodiversity conservation, and sustainable development.

Course Objectives (COs)

After successful completion of this course, students will be able to:

1. Develop a comprehensive understanding of cell biology, genetics, plant breeding, and evolutionary principles underlying plant growth, reproduction, inheritance, and variation.
2. Explain the fundamental principles and applications of biotechnology in agriculture, industry, environmental management, plant improvement, and sustainable resource utilization.
3. Develop an understanding of ecological processes, environmental issues, ecosystem functioning, and biodiversity conservation, with emphasis on sustainable management of plant resources.
4. Apply knowledge of plant physiology and phytochemistry to explain plant growth, metabolism, productivity, adaptation, and the importance of major plant biomolecules

Course Learning Outcomes (CLOs)

CLO	Course Learning Outcome (CLO)	Bloom's Taxonomy Level
CLO1	Explain the ultrastructure of plant cells, chromosome organization, and the mechanisms and significance of mitosis and meiosis.	Understand
CLO2	Explain the fundamental principles of inheritance, linkage, crossing over, mutation, and their roles in plant genetics and crop improvement.	Understand
CLO3	Describe the principles and applications of plant breeding, artificial pollination, selection methods, and theories of evolution.	Understand
CLO4	Explain the basic concepts and applications of biotechnology, including microbial fermentation, dairy biotechnology, and biogas production.	Understand
CLO5	Analyze ecological principles, ecosystem structure and function, ecological adaptations, biogeochemical cycles, environmental pollution, and greenhouse effects.	Analyze
CLO6	Explain the importance of biodiversity conservation and evaluate in-situ and ex-situ conservation strategies for sustainable management of	Evaluate

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	plant resources.	
CLO7	Explain the physiological processes of plants, including water relations, mineral nutrition, photosynthesis, respiration, enzyme action, photoperiodism, vernalization, and plant hormones.	Understand
CLO8	Describe the classification, structure, biological functions, and economic importance of major plant carbohydrates and communicate scientific concepts effectively through assignments and examinations.	Apply

Mapping of CLOs with PLOs

CLOs	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	√	√										
CLO2			√	√								
CLO3					√	√	√					
CLO4								√	√	√		
CLO5										√	√	√
CLO6	√	√	√									
CLO7				√	√							
CLO8						√	√					

Course Content

Unit Title	Detailed Unit Contents	Teaching-Learning Strategies	Assessment	CLO
Unit 1. Cytology Weeks 1–4	Ultra-structure of a eukaryotic cell, differences between prokaryotic and eukaryotic cells, physical and chemical composition of cell wall and plasma membrane. **Chromosome:** Physical and chemical nature, chromosome organization, types of chromosomes and role in heredity. **Cell division:** Mitosis and meiosis, stages and significance in growth, reproduction, genetic variation and evolution. **Cell organelles:** Structure and functions of nucleus, mitochondria, chloroplast, endoplasmic reticulum, ribosome, Golgi apparatus, vacuole and other cellular components.	Interactive lectures, multimedia presentations, classroom discussion, diagrams	Quiz, class test, assignment, final examination	CLO1
Unit 2. Genetic Weeks 5–8	**Basic concepts of genetics:** Gene, DNA and RNA structure, DNA replication, gene expression,	Lectures, multimedia presentations	Quiz, assignment, midterm,	CLO1, CLO2

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	Mendel's laws; linkage and crossing over; mutation, point mutation, physical and chemical mutagens and significance.		final examination	
Unit 3. Plant Breeding and Evolution Weeks 9–11	**Plant Breeding and Evolution:** Definition and significance of plant breeding. Method of artificial pollination in plants. Selection method. **Theories of evolution:** Darwin's theory and synthetic theory of evolution.	Lectures, case studies, group discussion	Class presentation, assignment, written examination	CLO2
Unit 4. Biotechnology Weeks 12–15	Definition, concept, history and scope of biotechnology; applications in agriculture, industry, medicine and environmental management. **Milk products:** Techniques of production of curd and butter; role of microorganisms in dairy fermentation. **Biogas technology:** Production methods, microbial role, composition of biogas and its uses as renewable energy. **Plant tissue culture and its applications:** Principles, techniques, micropropagation and importance in crop improvement and conservation. **Biofertilizers and biopesticides:** Types, production, applications and importance in sustainable agriculture.	Interactive lectures, demonstrations, case studies	Quiz, assignment, final examination	CLO2
Unit 5. Plant Ecology and Environmental Science Weeks 16–20	Definition, differences between autecology and synecology. **Ecological factors:** Climatic, edaphic and biotic. **Ecological adaptation:** Characteristics of hydrophytes, xerophytes and halophytes. **Ecosystem:** Definition; pond ecosystem; food chain; food web; ecological pyramids. **Biogeochemical cycles:** Nitrogen and carbon cycles. **Pollution:** Air and water pollution. **Greenhouse effects:** Definition, sources of greenhouse gases; causes	Lectures, field examples, case studies, group discussion	Assignment, presentation, midterm, final examination	CLO3

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	and effects of greenhouse gases; CO_2, CFC, CH_4 and N_2O; ozone layer depletion; carbon dioxide and the world climate; control of greenhouse effects.			
Unit 6. Conservation Biology	**Conservation:** Importance of conservation of plant species. Methods of ex-situ and in-situ conservation.	Interactive lecture, case studies	Assignment, written examination	CLO3
Unit 7. Plant Physiology and Phytochemistry Weeks 21–28	**Water relation:** Mechanisms of absorption of water and minerals. **Mineral nutrition.** **Photosynthesis:** General account and modern concept. **Respiration:** Aerobic and anaerobic. **Enzymes:** Definition and properties. **Photoperiodism.** **Vernalization:** Definition and mechanism. **Plant hormones:** A general account. **Carbohydrates:** Classification with examples; structures of important monosaccharides and oligosaccharides and isomerism; storage polysaccharides; starch, inulin, glycogen; structural polysaccharides; cellulose and economic importance of carbohydrates.	Lectures, tutorials, problem-solving, multimedia presentations	Quiz, class test, assignment, midterm, final examination	CLO4, CLO5

Course Assessment Framework:

Total Course Marks: 100

Assessment Category	Assessment Component	Weightage
Formative Assessment (20%)	Attendance & Class Participation	5%
	In-course Examination	10%
	Quiz / Assignment Presentation	5%
Summative Assessment (80%)	Final Examination • Students will have to answer 8 questions out of 12 • 10 marks per question ($8 \times 10 = 80$ marks) • Each question may contain maximum 3 sub-sections (e.g., a, b, c)	80%

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List of Books

Agrios, G. N. (1997). *Plant Pathology* (4th ed.). Academic Press, London, UK.

Lewin, B. (2000). *Gene 2000*. Oxford University Press & Cell Press, Oxford, UK.

Devlin, M. R., & Witham, H. F. (1986). *Plant Physiology* (4th ed.). CBS Publishers and Distributors, New Delhi, India.

Esau, K. (1953). *Plant Anatomy*. John Wiley & Sons, Inc., New York, USA.

Kumar, H. D. (1995). *General Ecology*. Vikas Publishing House, New Delhi, India.

Pelczar, M. J., Chan, E. C. S., & Krieg, N. R. (1993). *Microbiology: Concepts and Applications.* McGraw-Hill Book Company, New York, USA.

Vashishta, P. C. (1993). *Botany for Degree Students: Pteridophyta*. S. Chand & Company Ltd., New Delhi, India.

আখতারুজ্জামান, ম. (২০০০). *বিবর্তনবাদ*, হাসান বুক হাউজ, ঢাকা।

কর্মকার, যদুলাল. (২০০০) *উদ্ভিদ শরীর বিজ্ঞান*, হাসান বুক হাউজ, ঢাকা।

খন্দকার এম. (১৯৯৪). *ডলমনোলজী*, ঢাকা বিশ্ববিদ্যালয় প্রকাশনা, ঢাকা।

বাসার, এম. এ., এম.এ. হাসান এবং ম. রফিকুল ইসলাম. (২০০৪). *উদ্ভিদ বিজ্ঞান*, হাসান বুক হাউজ, বাংলা বাজার, ঢাকা।

হাসান, এম. এ. (১৯৯৬). *বাংলাদেশের ভেষজ উদ্ভিদ*, আশরাফিয়া বই ঘর, বাংলা বাজার, ঢাকা।

হাসান, এম. এ. এবং এম. কে. আলম. (১৯৯৭). *উদ্ভিদ শ্রেণী বিন্যাস তত্ত* (৩য় সংস্করণ), হাসান বুক হাউস, ঢাকা।

PROVISIONAL

Course Code	8223058	Marks:50	Credits: 2	Class Hours: 42
Course Title	Botany -II Practical			Exam Time: 6 hours

Course Description

This practical course develops fundamental skills in the identification, classification, and study of major plant groups, including algae, fungi, bryophytes, pteridophytes, gymnosperms, and angiosperms. It provides hands-on experience in specimen collection, dissection, mounting, microscopy, scientific drawing, plant physiology experiments, mitotic cell division, and ecological adaptation studies. The course emphasizes accurate observation, laboratory techniques, scientific documentation, specimen identification, and interpretation of botanical characteristics and their ecological significance.

Course Objectives (COs)

After completion of this course, students will be able to:

1. Develop practical skills in the identification, classification, and study of representative algae, fungi, bryophytes, pteridophytes, gymnosperms, and angiosperms through dissection, mounting, observation, and illustration.
2. Explain the morphological and anatomical characteristics of different plant groups and relate their structural adaptations to their ecological habitats.
3. Perform basic physiological experiments to understand fundamental plant processes such as plasmolysis, osmosis, and photosynthetic oxygen evolution.
4. Observe and identify different stages of mitotic cell division using plant materials and understand their significance in growth and development.
5. Develop skills in scientific observation, laboratory techniques, record keeping, scientific drawing, and interpretation of botanical specimens.

Course Learning Outcomes (CLOs)

CLO	Course Learning Outcome	Bloom's Taxonomy Level
CLO1	Identify and classify representative genera of algae, fungi, bryophytes, pteridophytes, gymnosperms, and angiosperms through practical study.	Understand
CLO2	Perform dissection, mounting, microscopic observation, scientific drawing, and description of plant specimens.	Apply
CLO3	Conduct basic plant physiology experiments and interpret the results related to water relations and photosynthesis.	Analyze
CLO4	Identify different stages of mitosis and explain their importance in plant growth and development.	Understand
CLO5	Analyze ecological adaptations of hydrophytic and xerophytic plants through morphological and anatomical observations.	Analyze
CLO6	Maintain laboratory records, prepare scientific reports, and demonstrate botanical knowledge through viva-voce.	Apply

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Mapping of CLOs with PLOs

CLOs	PLO 1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	√	√										
CLO2		√	√	√								
CLO3				√	√	√	√					
CLO4	√	√		√				√	√			
CLO5		√				√	√		√	√		
CLO6	√	√	√								√	
CLO7				√	√					√		
CLO8					√	√						

Course Content

Unit Title	Detailed Unit Contents	Teaching–Learning Strategies	Assessment	CLO(s)
Unit 1: Plant Diversity, Identification and Systematics	1. Detailed study including dissection (where necessary), mounting, drawing, description and identification with classification of the following genera: Algae: *Chlamydomonas, Oedogonium* and *Vaucheria*. Fungi: *Yeast* and *Ascobolus*. Bryophyte: *Riccia* and *Marchantia*. Pteridophyte: S*elaginella*. Gymnosperms: *Cycas*. Angiosperm: Poaceae and Fabaceae.	Demonstration of specimens, laboratory observation, microscopic study, dissection and temporary slide preparation, scientific drawing practice, guided identification exercises.	Practical examination, specimen identification, laboratory performance, scientific drawing, practical record assessment.	CLO1 CLO2
Unit 2: Identification of Plant Genera with Diagnostic Characters	2. Identification of the following genera with reasons: Algae: *Volvox*, *Polysiphonea* and *Fucas*. Fungi: *Agaricus*, *Puccinia* and *Penicillium*. Lichen: *Crustose*, *Foliose* and *Fructose*. Bryophyte: *Anthoceros*. Pteridophyte: *Psilotum*,	Specimen display, field-based observation, herbarium study, comparative analysis, identification keys and discussion.	Spot identification test, viva-voce, practical examination, herbarium/specimen report evaluation.	CLO1 CLO2

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	Azolla and *Pteris*. Gymnosperms: Male and female cones of *Cycas*. Angiosperms: Scientific names of common plants.			
Unit 3: Plant Physiology Experiments	Experimental study of basic physiological processes: **(a)** Demonstration of plasmolysis: principle, experimental procedure, observation and precautions. **(b)** Demonstration of osmosis: experimental setup, observation and interpretation. **(c)** Evolution of oxygen during photosynthesis: experimental demonstration and factors affecting photosynthetic activity.	Laboratory demonstration, hands-on experimentation, observation recording, data interpretation and discussion of physiological processes.	Experimental performance, observation note, practical report, viva-voce.	CLO3
Unit 4: Cytology and Cell Division	Study of different stages of mitosis in onion (*Allium cepa*) root tip: Preparation of root tip squash. Identification of stages: prophase, metaphase, anaphase and telophase. Significance of mitosis in plant growth and development.	Microscopic observation, slide preparation demonstration, image-based learning, identification and interpretation of mitotic stages.	Microscopic identification, slide preparation assessment, practical examination and viva.	CLO4
Unit 5: Plant Ecology and Adaptation Studies	**(a)** Morphological and anatomical studies of hydrophytic and xerophytic plants. **(b)** Ecological identification of plants: Identification of plants according to habitat. Scientific naming and ecological significance.	Field observation, specimen collection, ecological comparison, anatomical study, group discussion and field-based learning.	Field report, plant identification test, practical record evaluation, viva-voce.	CLO5
Unit 6: Laboratory Documentation and Scientific Communication	Preparation and maintenance of laboratory notebook including: Recording observations. Scientific drawings and labeling. Classification and description of specimens. Preparation of practical reports.	Guided note preparation, feedback on scientific documentation, demonstration of proper laboratory practices.	Laboratory notebook evaluation, report assessment, continuous assessment.	CLO6
Unit 7: Viva-	**Viva-Voce:** Assessment of	Interactive	Final practical	CLO6

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Voce and Practical Competency Evaluation	students' understanding of: Plant identification and classification. Laboratory techniques. Experimental principles and observations. Ecological and cytological concepts.	questioning, specimen-based discussion, problem-solving and conceptual assessment.	examination, viva-voce, overall laboratory performance assessment.	

List of Books

Agrios, G.N. (1997). *Plant Pathology* (4th ed.). Academic Press, London.

Benjamin L. (2000). *Gene 2000*. Oxford University Press and Cell Press. Bold, H.C. and

Kumar, H.D. (1995). *General Ecology*, Vikash Pub. House, India.

Lawrence, G.H.M. (1951). *Taxonomy of Vascular Plants*. The Macmillan Co. New York.

Pelczer, M. J., Chan, E. C. S. & Krieg, N. R. (1993). *Microbiology: Concepts and Applications*. McGraw Hill Book Co. Inc. New York.

Vashista, P.C. 1993). *Botany for Degree Students: Pteridophyta.* S.C. Chand & Co. Ltd. Ramnagar, New Delhi.

আখতারুজ্জামান, ম. (২০০০). *বিবর্তন বাদ*, হাসান বুক হাউজ, ঢাকা।

কর্মকার, যদুলাল. (২০০০). *উদ্ভিদ শরীর বিজ্ঞান*, হাসান বুক হাউজ, ঢাকা।

খন্দকার মনিরুজ্জামান, (১৯৯৪). *লিমনোলজী*, ঢাকা বিশ্ববিদ্যালয় প্রকাশনা, ঢাকা।

বাসার, এম. এ., এম.এ. হাসান এবং ম. রফিকুল ইসলাম. (২০০৪). *উদ্ভিদ বিজ্ঞান*, হাসান বুক হাউজ, বাংলা বাজার, ঢাকা।

হাসান, এম. এ. (১৯৯৬). *বাংলাদেশের ভেষজ উদ্ভিদ*, আশরাফিয়া বই ঘর, বাংলা বাজার, ঢাকা।

হাসান, এম. এ. এবং এম. কে. আলম. (১৯৯৭). *উদ্ভিদ শ্রেণী বিন্যাস তত্ত* (৩য় সংস্করণ), হাসান বুক হাউস, ঢাকা।

PROVISIONAL

Instruction to the Examiners

Time: 6 hours Marks: 50

1. Detailed Study, Dissection, Mounting, Description and Identification of Plant Genera

Marks: 12 (Slide preparation/mounting: 2+ Neat labeled diagrams: 2+ Identification: 0.5+ Classification and identifying characters: 1.5=6×2=12)

Specimen A and B:
Two specimens will be provided from the following groups:

- Algae: *Chlamydomonas, Oedogonium, Vaucheria*
- Fungi: Yeast, *Ascobolus*
- Bryophyta: *Riccia, Marchantia*
- Pteridophyta: *Selaginella*
- Gymnosperms: *Cycas*
- Angiosperms: Families Poaceae and Fabaceae

Students will perform:

- Dissection (where necessary)
- Preparation of slides/mounting
- Observation and identification
- Classification
- Description of diagnostic features
- Scientific drawings with proper labeling

2. Spot Identification of Plant Genera with Reasons Marks: 9

(Identification: 0.5+ Identifying Characteristics: 1.0=1.5×6=9)

Specimens C–H will be provided for identification from the following groups:

Algae: *Volvox, Polysiphonia, Fucus*; Fungi: *Agaricus, Puccinia, Penicillium*

Lichens: Crustose, Foliose, Fruticose; Bryophyta: *Anthoceros*

Pteridophyta: *Psilotum, Azolla, Pteris*

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Gymnosperms: Male and female cones of *Cycas*;

Angiosperms: Common flowering plants with scientific names

3. Plant Physiology Experiments Marks: 6

(Experimental setup and procedure: 3+ Observation and interpretation: 2 + Precautions and conclusion: 1= 6)

Students will perform experiments related to:

(a) Demonstration of Plasmolysis: Principle, Experimental procedure, Observation, Precautions

(b) Demonstration of Osmosis: Experimental setup, Observation and interpretation

(c) Evolution of Oxygen During Photosynthesis: Experimental demonstration, Observation of oxygen release, Factors affecting photosynthetic activity

4. Cytology: Study of Mitosis in Onion Root Tip Marks: 6

(Preparation of slide: 2+ Identification of mitotic stages: 2+ Neat labelled diagrams: 1+ Significance/comment:1=06)

Students will prepare and study onion (*Allium cepa*) root tip squash.

Stages to be studied: Prophase, Metaphase, Anaphase, Telophase

5. Plant Ecology and Adaptation Study Marks: 4

Specimen I and J (Identification:0.5+ Identifying characters/adaptive features:1.5=2×2=4)

Specimens will be selected from:

- Hydrophytic plants
- Xerophytic plants

Students will study:

- Morphological adaptations, Anatomical adaptations
- Ecological significance, Identification of plants with scientific names

6. Collection, Laboratory Record and Viva-Voce Marks: 13

Components	Marks
Collection of plant specimens (Algae, Fungi, Bryophyta, Pteridophyta, Gymnosperms and Angiosperms)	3.0

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Components	Marks
Practical Laboratory Notebook	5.0
Viva-Voce	5.0
Total	13.0

Total Marks Distribution

Examination Component	Marks
Detailed study of specimens (A & B)	12
Spot identification (C–H)	9
Plant physiology experiment	6
Cytology (Mitosis)	6
Ecology and adaptation study	4
Collection	3
Practical Notebook	5
Viva-Voce	5
Grand Total	50

General Instructions

- Students must submit properly prepared practical notebooks containing classification, descriptions, observations and labelled drawings.
- Slides must be prepared carefully following standard laboratory techniques.
- Identification answers must include appropriate diagnostic characters.
- Scientific names should be written according to accepted botanical nomenclature.
- Herbarium specimens and field collections must include proper labels.
- Viva-voce will evaluate students' understanding of practical techniques, specimen identification and botanical concepts.

PROVISIONAL

Course Code	**8223160**	**Marks: 50**	**Credits: 2**
Course Title	**Viva-voce**		

1. Presentation on field work report: 20
2. Viva on all courses of Zoology: 30